Most erosion stems from human actions

Inlets, shoreline armoring among major causes

By Kevin Lollar

Sand is notoriously shifty, unstable stuff. Scientists like to use the word "dynamic" to describe sandy environments, particularly beaches on barrier islands.

More simply put, natural forces — wind, waves, currents — move beach sand around a lot.

With all this movement, beach erosion happens naturally. But toss boats and beachfront development into the mix, and beaches really begin to disappear.

Barrier islands are, indeed, dynamic: Natural forces constantly change the shapes of beaches.

Wind, for example, blows sand shoreward to help build dunes; waves pull sand from the beach out to sea and push it back on shore.

Longshore currents, which run parallel to the beach like a watery conveyor belt, are another important sand mover: Sand from beaches is caught in longshore currents, moves along the shoreline and ends up on a beach somewhere down the coast.

In Florida, the general flow of longshore currents is north to south, so the sand generally moves from north to south, flowing across inlets from one barrier island to another.

This process is called "sand sharing." Barrier islands constantly lose sand to downdrift islands, but they constantly gain sand from updrift islands. Storms and high tides can come along and take big bites out of barrier islands, dragging sand offshore — usually, though, the sea puts that sand back onto the beach.

So, for the most part, in the natural world, beaches don't suffer incurable erosion.

That's the way it works until people become involved.

In blunt terms, Per Bruun, the father of beach preservation in Florida, wrote in "The History and Philosophy of Coastal Protection" that "Man-made erosion is a black spot on man's association with shores. It is a deplorable fact that all coastal protective measures apart from artificial nourishment (may) have an adverse affect on adjoining shores."

The human factor

Let's start with navigational inlets.

Many people who live or vacation along the coast are boaters, and many of those boaters want to take their vessels into the Gulf of Mexico or Atlantic Ocean.

To get there, they need deep passage between barrier islands. But most natural inlets in Florida are too shallow for safe navigation, so the U.S. Army Corps of Engineers periodically dredges them.

These navigational inlets, though, are the major cause of beach erosion in the state.

Since most inlets are relatively shallow, gentle tidal currents flow through them and sand sharing can take place.

When inlets are dredged for navigation, the deeper channels have swifter currents, so instead of moving to the next island, sand in longshore currents is blown out to sea on outgoing tides or sucked into the estuary on incoming tides.

Beaches next to inlets can be erosion hot spots, so the Corps of Engineers often builds jetties — rock structures running parallel to the shoreline — next to passes.

Jetties prevent erosion by capturing sand in the longshore current and the beach on the updrift side of the jetty actually builds up.

But jetties prevent sand from flowing to the next island, which is losing sand naturally in the longshore current, and the downdrift island erodes. This is called "sand starvation."

In a recent paper on inlet-induced erosion, state coastal engineer Emmett Foster lists 24 inlet areas that show downdrift erosion, including Sebastian, Lake Worth, Boca Raton and Jupiter on the Atlantic coast and Clearwater, Gasparilla and Big Marco on the Gulf coast.

Because of its role in maintaining navigational inlets, the Corps of Engineers has been singled out as a main cause of beach erosion.

The Corps is getting a bum rap, said Stephen Leatherman, director of the Laboratory for Coastal Research and International Hurricane Center at Florida International University.

"The Army Corps comes in at local, county and state request," Leatherman said. "No one perceived the problems of navigational inlets early on. Now it's obvious to everybody. People were thinking, 'Hey, we want to get our boats out.' It was a way to pump up the economy. The Corps was a willing partner, but people want to get their boats out. You can't blame the Corps."

Inlet solutions?

University of Florida coastal engineering professor Robert Dean says there's good news about human-caused erosion.

What man has done, he can undo.

Inlets carved into the Florida coastline during the past century or so account for 80 percent to 85 percent of the erosion on the Atlantic coast and about 35 percent on the Gulf coast.

Dean said the answer lies in investing in sand bypassing systems that transport sand from one side of the inlet to another. Currently, two of Florida's inlets have permanent bypassing plants and two others have floating dredges dedicated to bypassing sand.



Shoreline armoring

Beachfront development is the other big human cause of erosion.

"Without buildings, there would be less of a public need to do much about it," Foster said. "When you place a building in harm's way, it becomes a public need to manage erosion."

Consider a home built on a beach. Longshore currents remove sand from in front of the building, and the homeowner begins to wonder about the future of his or her home. Or, more startling, a storm hauls off a big chunk of the homeowner's beach. Instead of waiting for natural processes to replace the sand, the panicky homeowner builds a seawall in front of the house.

Now consider a larger section of eroding beach where the sea threatens many houses or the tourist industry.

Officials then put a groin — the same thing as a jetty, only not on the end of the island — at the downstream side of the eroding section of beach. Coastal scientists call building jetties, seawalls and groins armoring or hardening the shoreline. They also call the strategy a bad idea.

A seawall protects the house, but waves scour sand away from in front of the sea wall; the beach, therefore, can't build up, and eventually nothing is left in front of the house but the seawall itself.

Like jetties, groins let sand build up on the updrift side but don't let it drift naturally to the downstream side. So the downdrift beach suffers sand starvation.

When that happens, people on the sand-starved beach put in their own groin, which starves the next section of beach and prompts another groin, and then another, until miles of beach become a vast and unattractive groin field.

Armoring shorelines used to be the way to preserve beaches, but it has fallen into disrepute because, while armoring might save private property, it often destroys public beach.

"If you had the home, I don't think you'd say it's bad," Bruun said. "In an emergency, armoring is good for those sitting behind it, but it's not good for the neighbors. It's hard to say, 'Look here: You are supposed to protect your property, but you shouldn't do damage to the Smiths and the Joneses.' "

Beach nourishment

These days, more communities and coastal scientists are using a more sand-friendly way to preserve beaches: Beach nourishment rebuilds beaches by pumping sand onto the shoreline from offshore or trucking it in from inland sites. But nourished beaches can't stand up to the forces of man and nature forever, and they, too, erode.

"Communities have turned to nourishment as a panacea," Leatherman said. "I agree with nourishment. Without that recreational border, you're not going to get the tourists. Nobody wants to come sit on a seawall and look the ocean.

"We're between a rock and a hard place. I just hope we're not between the ocean and a seawall. In that case, the beach will be squeezed out."

Beach nourishment is not a panacea. Nothing is.

Even if all navigational inlets were filled and all armoring removed, and every beach were pumped full of sand, global warming would cause slow, steady beach erosion.

Global warming and global cooling are nothing new. During the last great ice age, a vast amount of the Earth's water was frozen and sea levels dropped 300 feet — Florida's Gulf-front beaches were more than 100 miles west of where they are now.

As the Earth warmed, sea levels rose. The Earth is still warming, the seas are rising and the rising water erodes beaches.

During the past 100 years, sea levels have risen an average of 6 inches worldwide. During this century, the rate of sea level rise could be three or four times greater, Leatherman said.

That amount of sea level increase could cause beaches to erode as much as 200 feet.

"I'd hate to see that happen — I love beaches," Leatherman said. "I certainly hope the forecasts are wrong, but we're going to see more erosion. I'd like to believe there's a magic bullet out there, but until the Corps of Engineers finds a plug to the bottom of the ocean so they can let some water out, we've got a problem."

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Michael R. Brown, FLORIDA TODAY

Blocking the flow: In 1954, the U.S. Army Corps of Engineers completed a 400-foot wide channel and two jetties at what would become Port Canaveral. Shortly after completion, the Corps admitted the inlet blocked the normal north-to-south flow of sand, robbing beaches to the south. Inlets on Florida's Atlantic coast are responsible for 80 percent to 85 percent of all beach erosion.

Preservation pioneer crunched numbers, saved beaches

By Kevin Lollar

JFK had a problem. Jackie had a new bathing suit, but no place to wear it — erosion had destroyed the beach by the Kennedy compound at Palm Beach. So the call went out for the one man in the state who could save the day: Per Bruun, Florida's father of beach preservation.

Bruun's solution was to nourish the beach by increasing the efficiency of a sand transfer plant, which was not doing a good job of pumping sand across a nearby inlet.

"At high tide, there was actually no beach, and I was called in to do the dirty work," said Bruun, 85. "Everything worked out: Jacqueline got her new bathing suit, President Kennedy got his new beach, and I got a new wave generator."

Four decades later, Bruun remains active in semi-retirement. He recently took time on a breezy and cool South Carolina day to reflect on his long, remarkable career, show off his island and tell a lot of very funny stories.

Curiously, if not for Hans Albert Einstein Jr., son of the famous physicist, Jackie Kennedy might never have gotten to lounge on the beach in her new bathing suit.

During the early 1950s, Florida's political leaders realized many beaches statewide were eroding badly and asked University of Florida scientists to find someone who could do something about it. The university turned to Einstein at the University of California-Berkeley, who recommended Danish engineer Bruun.

The two scientists had met in 1952 when Bruun was a Fulbright Scholar at Berkeley.

"Hans Albert was a German Jew, and I had an advantage because I could speak German, so we became friends," Bruun said. "He liked me because I introduced mathematics to coastal geomorphology. He was a mathematician, and any time I came up with mathematics, he was loving me."



Craig Rubadoux, FLORIDA TODAY

> "Beaches in Florida were in bad shape. I could see what was happening, and I said, 'Look here: How can we handle this?' "
>
> **PER BRUUN,** coastal engineering pioneer

On Einstein's recommendation, the state hired Bruun, who arrived in Gainesville with his wife and two children Dec. 13, 1954. Bruun set to work on Florida's erosion problems, founding and becoming the first chairman of the University of Florida's Department of Coastal Engineering.

He organized the Florida Shore & Beach Preservation Association, a "league of coastal cities and counties" that lobbies for beach preservation programs.

Through the association, he helped persuade the Legislature in 1957 to create the State Department of Beach and Shore Erosion, which authorized and shared the cost of beach projects. The department later became the Florida Division of Beaches and Shores, and is now the Bureau of Beaches and Wetland Resources.

Early advocate

Early on, Bruun became an advocate of beach nourishment — the process of putting sand on eroding beaches.

The 10th of Bruun's "Ten Demands for Coastal Protection," which he calls "as strict and demanding as those listed in Deuteronomy," sums his theory of nourishment: "Thou shalt show forgiveness for the sins of the past and cover them up in sand."

As a mathematician, Bruun viewed erosion in terms of equations, an approach that some scientists didn't buy.

"Beaches in Florida were in bad shape," Bruun said. "I could see what was happening, and I said, 'Look here: How can we handle this? Let's try to put this in mathematical terms.' At that time, people shook their heads and said, 'This guy's a little bit crazy.' Look here: If a man believes he can put everything in mathematical terms, most people will say he's odd."

Stephen Leatherman, director of the Laboratory for Coastal Research and International Hurricane Center at Florida International University, confirms some scientists thought Bruun was a little bit crazy.

"He was a man way ahead of his time," Leatherman said. "He came up with the Bruun Rule, a fairly simple volumetric equation to show rising sea level actually causes erosion. At the time, people said, 'No, it's not possible. There's no relationship.' We've done a lot of work since then that has pretty much showed his work is correct."

Bruun succinctly stated his solution to the problem of erosion caused by sea level rise in a 1972 paper called "The History and Philosophy of Coastal Protection."

"It is obvious that the only way in which this erosion can be counteracted is by artificial nourishment replacing the material eroded by other material, whether from land or from offshore sources," he wrote.

Bruun began thinking about beaches as a young boy in Skagen, the northernmost city in Denmark, which sits on the coast of the often-tempestuous North Sea.

"I come from the roughest shores in the world," he said. "You talk about 10- or 15-foot waves here. That doesn't impress me. Look here: I'm used to 30-, 40-, 50-foot waves, up to 32 meters — that's more than 100 feet. They are mountains of water. They are monsters. You can't really imagine it. You have to see it. You talk about losing hundreds of thousands of cubic yards of sand in a storm; I'm talking about losing millions.

"As a kid, I was deeply impressed with these fantastic natural forces. Though I didn't understand them, I felt something was there. I had a boat when I was very young because I had an idea very early that I'd become an engineer working with the sea. I started working with mathematics fairly early, and by the time I was 23 years old, I said, 'Look here: I'd like to understand what's happening, and the best thing is to put it into an equation.' "

During the late 1970s, another call went out for Bruun's help: The Hilton Head (S.C.) Chamber of Commerce needed somebody to fix the island's badly eroded beach.

"They were in despair," Bruun said. "It was a terrible situation. I'm the one who kept this island afloat."

This time, Bruun's solution was to have 15 miles of canals dug on the island and use the 1.6 million cubic yards of dredged sand to nourish the beach. In 1979, Bruun went into semi-retirement and moved with his wife, Elizabeth, permanently to Hilton Head.

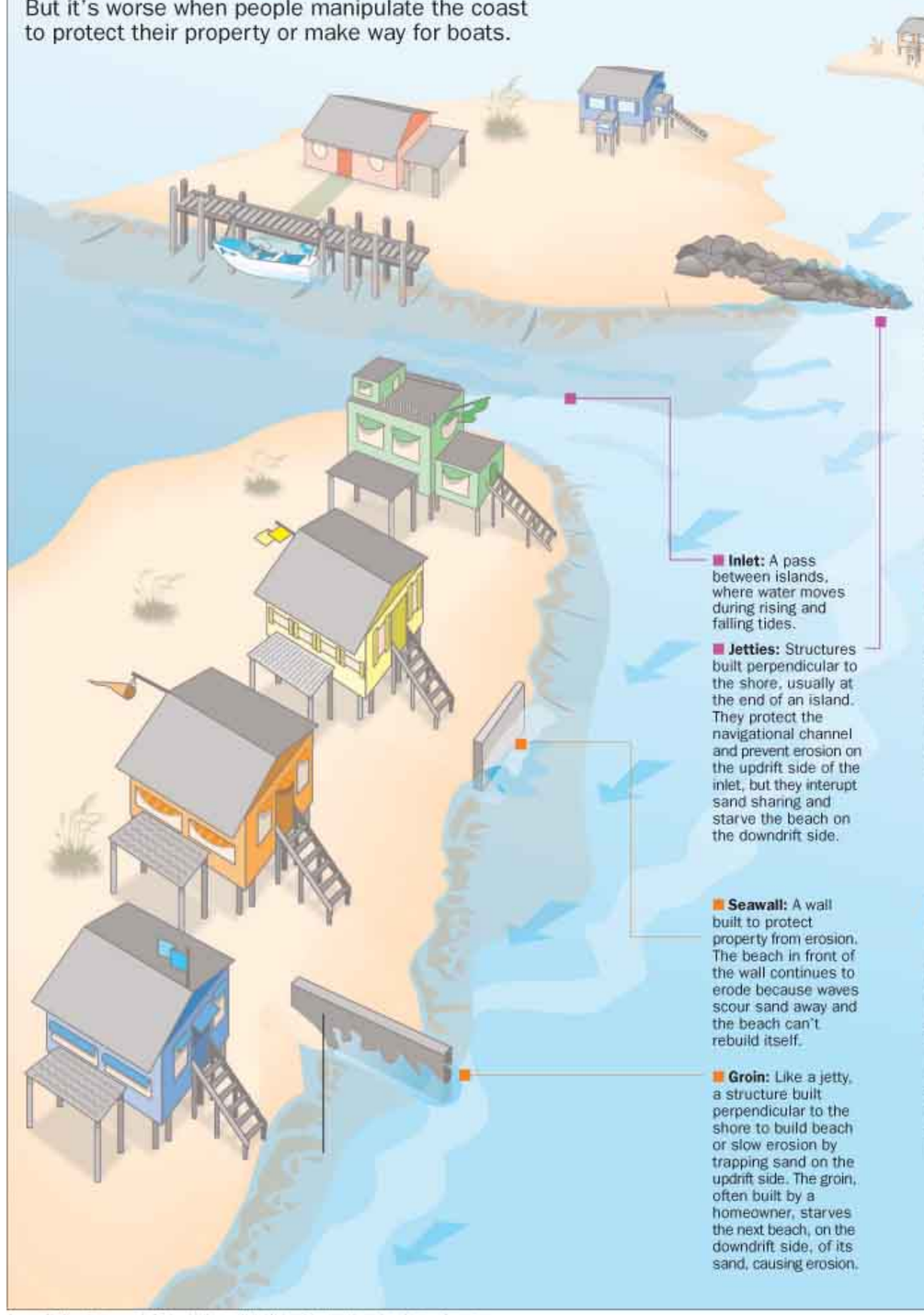
"My wife and I are both northern by birth," he said. "This is a better climate for us."

Now, at the age of 85, after heart and cataract surgery, Bruun stays active, doing research, consulting, writing, and lecturing.

"I just try to keep up as good as I can. Physically, I'm not a young man, but there's nothing wrong with my mind," he said.

What causes beach erosion?

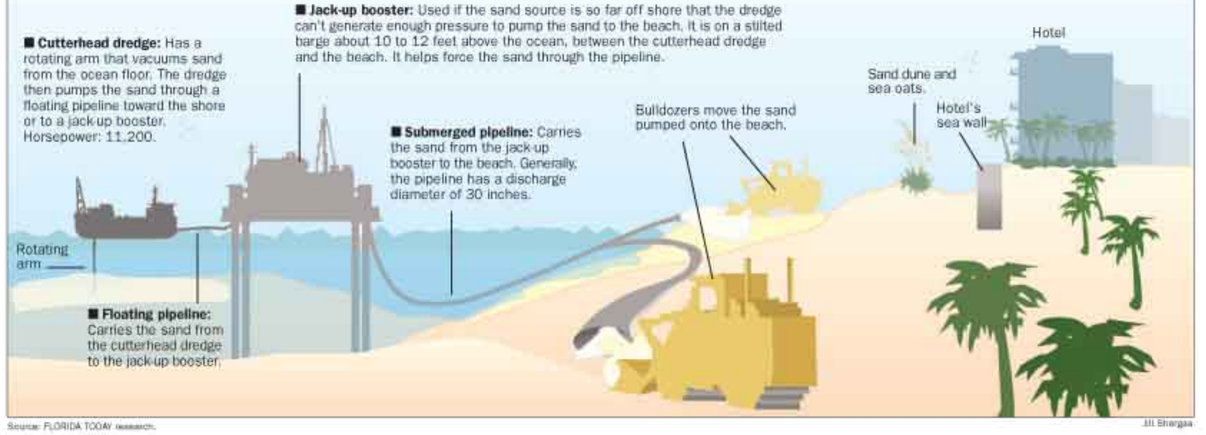
Erosion happens naturally.
But it's worse when people manipulate the coast to protect their property or make way for boats.



Sources: The News-Press research; Stephen Leatherman; United Nations Educational Scientific and Cultural Organization — Jill Sheridan

Beach renourishment process

Sand is brought in from offshore and deposited at a rehandling area.



Source: FLORIDA TODAY research. — Jill Sheridan